object.cnt

Purpose

`object.cnt` declares the basin name, the landscape and total areas, and how many of each spatial object type (HRU, channel, aquifer, reservoir, and so on) the project contains. The model uses these counts to allocate the `ob`, `obcs`, `obcs_alloc`, and `obom` arrays before any object data is read.

If `object.cnt` is missing or set to `null` in `file.cio`, the model prints `Cannot find object.cnt input file` and stops.

Source

- Reader: src/basin_read_objs.f90
- Type definitions:
  - src/basin_module.f90, `type basin_inputs` (`bsn`)
  - src/hydrograph_module.f90, `type spatial_objects` (`sp_ob`)

Format

- Line 1: title line. Skipped.
- Line 2: header line. Skipped.
- Line 3: 21 values read into `bsn` and `sp_ob`, list-directed.

#	Field	Maps to	Type	Description
1	name	`bsn%name`	char(25)	basin name (free-form)
2	ls_area	`bsn%area_ls_ha`	real	landscape area, hectares
3	tot_area	`bsn%area_tot_ha`	real	total area, hectares
4	objs	`sp_ob%objs`	int	total object count (sum of all object types)
5	hru	`sp_ob%hru`	int	number of HRUs
6	hru_lte	`sp_ob%hru_lte`	int	number of HRU-lte objects
7	ru	`sp_ob%ru`	int	number of routing units
8	gwflow	`sp_ob%gwflow`	int	number of gwflow cells
9	aqu	`sp_ob%aqu`	int	number of aquifer objects
10	chan	`sp_ob%chan`	int	number of legacy channels
11	res	`sp_ob%res`	int	number of reservoirs
12	recall	`sp_ob%recall`	int	number of recall objects
13	exco	`sp_ob%exco`	int	number of export-coefficient objects
14	dr	`sp_ob%dr`	int	number of delivery-ratio objects
15	canal	`sp_ob%canal`	int	number of canals
16	pump	`sp_ob%pump`	int	number of pumps
17	outlet	`sp_ob%outlet`	int	number of outlets
18	chandeg	`sp_ob%chandeg`	int	number of SWAT-deg channels (SWAT+ routing)
19	aqu2d	`sp_ob%aqu2d`	int	2D aquifer count (currently not used per the type comment)
20	herd	`sp_ob%herd`	int	herd count (currently not used per the type comment)
21	wro	`sp_ob%wro`	int	water-rights object count (currently not used per the type comment)

gwflow side effect

After reading, if `bsn_cc%gwflow == 1` (set via `codes.bsn`), the reader looks for `gwflow.chancells`. If that file exists and contains river cells, the reader:

- sets `sp_ob%gwflow` to the number of river cells,
- recomputes `sp_ob%objs` to `sp_ob%objs + nriv - sp_ob%aqu`,
- sets `in_con%gwflow_con` to `gwflow.con`,
- sets `sp_ob%aqu` to 0 and `in_con%aqu_con` to `null`.

If `bsn_cc%gwflow == 1` but `gwflow.chancells` is missing or empty, gwflow is silently turned off.

Example

`refdata/Ames_sub1/object.cnt`:

```
object.cnt:
name                    ls_area      tot_area       obj       hru      lhru       rtu       mfl       aqu
demo                       1.           1            12         12         0         0         0         0
```

Notes on the example:

- `name = demo`, `ls_area = 1` ha, `tot_area = 1` ha. The fields are floats; the trailing `.` makes that explicit in this dataset.
- `objs = 12` and `hru = 12`. Every spatial object in this project is an HRU.
- The header uses shortened labels (`obj`, `lhru`, `rtu`, `mfl`, `dlr`, `can`, `pmp`, `out`, `lcha`, `hrd`). These are tool conventions; the reader does not parse the header.

Related files

- `codes.bsn`. Sets `bsn_cc%gwflow`, which triggers the gwflow side effect above.
- `file.cio`. The `simulation` category points to `object.cnt`.
- Object-specific connectivity files (`hru.con`, `chandeg.con`, `aquifer.con`, `reservoir.con`, …). These must agree with the counts in `object.cnt`.